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Proposal for:
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Proposal Number:
QU-0095

Reference
US WI WINDWARD GLITC

Understanding of project

Windward has requested for pricing on design assistance for a ground source heat pump (GSHP) system project located on Pine Square Rd in Cassian, Wisconsin. The building is a juvenile center and is approximately 35,000 ft^2. GEOptimize will work with the engineer of record (Windward) to produce a detailed ground heat exchanger (GHX) design as well as assist in equipment selection.

Proposed design methodology

Accurate hourly energy load profiles of the buildings connected to a GHX are critical to the development of a GSHP system that is cost-effective and performs efficiently and predictably for the life of the building. Energy loads are determined by simulating building energy models based on the architectural and mechanical system design, anticipated occupancy, and the historical hourly weather data of the location.

GEOptimize will review an hourly energy model based on drawings, specifications, and for the project provided by Windward. The energy model will be developed in Trane Trace 700 specifically for the outputs required to import into Ground Loop Design (GLD2016) to model the size and performance of a proposed GHX. GEOptimize will request a .GL1 file to be outputted from the Trane Trace 700 software. Iterations of the energy model *may* be recommended to determine the impact of several energy efficiency measures on the annual energy load balance and the proposed GHX design. Some of the iterations *may* include:

- Alternate glass specifications with different U-values and solar heat gain coefficients to modify cooling and heating energy and peak loads.
- Calculation and inclusion of domestic hot water to reduce heat rejection to the GHX
- Additional heating load for areas of snowmelt on the site
- Exhaust air energy recovery system to reduce overall heating and cooling loads

- Other discretionary heating and / or cooling loads that may become apparent during design development

The information will be used to create a set of iterative hourly energy models that may be used to inform the design with recommendations of changes that can potentially be made to the building and / or mechanical systems that will result in a reduction in the size / cost of the GHX needed to service the building loads.

Thermal properties of the soil and rock at the site will be originally estimated from geotechnical reports or information available online. A thermal conductivity test will be coordinated to verify the assumptions once the design is proven technically and economically feasible for the client. Consideration is given to all potential GHX designs and configurations based on building energy loads, geology, land area available for construction of the GHX and on the capabilities of local GHX contracting firms. Potential GHX designs may include vertical borehole configurations, horizontally excavated or horizontally bored GHX fields as well as groundwater options.

Analysis of building energy and GHX models will often indicate the amount of heat rejected to the GHX is significantly different than the amount of heat extracted. This implies the energy loads to and from the GHX are unbalanced. Using energy modeling to inform the design can significantly improve the energy balance to and from the GHX. This will reduce the size and cost of the GHX while helping ensure long-term efficient performance.

Analysis of GHX models developed based on the information may indicate the amount of energy that must be extracted from the ground to heat the buildings is unequal to the amount of heat that will be rejected to it while cooling the buildings. Additional iterations of the building energy models will be developed to determine the impact of a range of energy efficiency measures on the energy balance and the size and cost of the GHX. Balancing the energy loads:

- Reduces the size of the GHX, reducing first cost
- Avoids long-term temperature increase or decrease in the ground that affects the long-term viability of the system, and affects system efficiency and energy cost
- Creates a system that operates predictably into the future

Scope of work

Energy load profiles

Detailed hourly energy models will be developed by Windward Engineers & Consultants using Trane Trace 700 based on drawings, specifications and information for each building provided by Windward. The energy model developed will be specifically used to produce a .GL1 file for importing into GLD2016. GEOptimize will review the energy model and recommend changes as necessary.

Detailed GHX design

Details of the GHX sketches provided will include:

- Layout of GHX circuit piping for the recommended GHX configuration
- Design of fusion welded headers to connect GHX circuits to supply / return runout piping
- Supply / return runout piping from GHX headers to manifold in the building
- Specifications for heat transfer fluid and estimated heat transfer fluid volume
- GHX manifold design up to isolation valves between heat pump system piping and GHX
- Pressure-drop calculations for the GHX and recommended circulation pump sizing and control

- Recommendations for heat pump selection/sizing
- Assist in selecting the GHX contractor
- Assist in developing the conceptual sequence of operations.

Long-term system monitoring (recommended)

To ensure efficient operation of the GSHP system, GEOptimize recommends continuous monitoring of the energy loads to and from the GHX and the temperature of the heat transfer fluid delivered to the heat pumps connected to the system. GeoFease has developed a cloud-based energy monitoring system designed to:

- Monitor and log fluid temperatures and flows through the GHX continuously.
- Calculate energy transfer between the ground and building and log the data on a remote server.
- Display the information about the GHX performance on a dedicated, password protected website for the project.
- Predict the performance of the GHX into the future based on measured energy loads to and from the GHX and the design of the GHX.
- Provide the building owner/operator advance notification if the performance of the GHX is trending outside of specified flow rates and temperatures.
- Optionally use the predictive monitoring capability of the system to transmit a signal to the building automation system to enable auxiliary heating and/or cooling devices to manage the performance of the GHX and eliminate the potential for long-term temperature degradation.

Long-term monitoring is available on a subscription basis for the end user. Please request pricing at https://geofease.com/contact/.

Deliverables

GEOptimize will prepare a PowerPoint presentation with sketches to be delivered to Windward at a meeting via screen sharing software detailing the conceptual system design. The construction drawings will be provided to the engineer of record as sketches. GEOptimize will work with the engineer of record to ensure a proper design is achieved and that they are confident with the assumed liability of signing off on the system.

Pricing

Description	Rate	Qty	Line Total
Detailed Design Assistance	$21,500.00	1	$21,500.00
Monitoring Consulting	$2,000.00	1	$2,000.00
	Subtotal		$23,500.00
Proposal Total (USD)			$23,500.00

Terms

- Additional work out of scope will be charged at $250/hr.
- Invoices will be submitted monthly for work completed to date. Payment is expected on receipt of invoice.

Authorized signature – Windward